

Mathematics Practice Test 02 (Adaptive)

Format: ISC Subjective

Portion: Set 1 Midterm

Time Allowed: 2 Hours

Maximum Marks: 60

General Instructions:

- All questions are compulsory. Internal choice is provided in Section D.
 - **CRITICAL RULE:** Enforce strict mathematical presentation. Always define variables explicitly (e.g., "Let $x = \dots$ "), do not drop limit or dx/dy notations, and state formulas before substituting values.
 - **NO BLANKS ALLOWED:** A -1 mock penalty will be applied to your final score for any question left completely blank.
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SECTION A (10 Marks)

Answer all questions. Each question carries 1 mark.

Question 1: If A is a square matrix of order 3×3 such that $|A| = 4$, find the value of $|\text{adj } A|$.

Question 2: The greatest integer function $f(x) = [x]$ is continuous at: (a) $x = 1$ (b) $x = -2$ (c) $x = 1.5$ (d) $x = 0$

Question 3: Find $\frac{dy}{dx}$ if $y = \log_e(\cos x)$.

Question 4: Write the direction cosines of a vector that is equally inclined to the positive directions of the axes.

Question 5: If A and B are independent events with $P(A) = 0.3$ and $P(B) = 0.4$, find $P(A \cap B)$.

Question 6: State the domain of the function $y = \sec^{-1}x$.

Question 7: If $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$, find $A + A^T$.

Question 8: Evaluate: $\int e^x (\sin x + \cos x) dx$ (if integral calculus is introduced, else skip/treat as basic derivative application. Wait, let's substitute with a strict Set 1 question). *Correction for Q8:* Write the condition for two lines with direction ratios (a_1, b_1, c_1) and (a_2, b_2, c_2) to be perpendicular.

Question 9: A matrix has 12 elements. What are the possible orders it can have?

Question 10: Find the unit vector in the direction of $\vec{a} + \vec{b}$, where $\vec{a} = 2\hat{i} - \hat{j} + 2\hat{k}$ and $\vec{b} = -\hat{i} + \hat{j} - \hat{k}$.

SECTION B (14 Marks)

Answer all questions. Each question carries 2 marks.

Question 11: Evaluate the principal value of: $\sin^{-1} \left(\sin \frac{2\pi}{3} \right)$.

Question 12: Without expanding, prove that the determinant of a skew-symmetric matrix of odd order is always zero.

Question 13: Find the area of a parallelogram whose adjacent sides are given by the vectors $\vec{a} = \hat{i} - \hat{j} + 3\hat{k}$ and $\vec{b} = 2\hat{i} - 7\hat{j} + \hat{k}$.

Question 14: Check whether the relation R on the set of real numbers $\mathbb{R}$ defined as $R = \{(a, b) : a \leq b^3\}$ is transitive. Justify.

Question 15: Differentiate $y = \tan^{-1}\left(\frac{\sin x}{1 + \cos x}\right)$ with respect to x by first simplifying the trigonometric expression.

Question 16: A die is thrown twice and the sum of the numbers appearing is observed to be 6. What is the conditional probability that the number 4 has appeared at least once?

Question 17: Find the scalar projection of the vector $\vec{a} = 2\hat{i} - 3\hat{j} + 6\hat{k}$ on the vector $\vec{b} = \hat{i} + 2\hat{j} + 2\hat{k}$.

SECTION C (21 Marks)

Answer all questions. Each question carries 3 marks.

Question 18: Differentiate $y = x^{\sin x} + (\sin x)^{\cos x}$ with respect to x . (Hint: Use logarithmic differentiation carefully on each term).

Question 19: Discuss the continuity of the function $f(x)$ at $x = 0$: $f(x) = \begin{cases} \frac{\sin 3x}{x}, & \text{if } x < 0 \\ \frac{e^{3x} - 1}{x}, & \text{if } x = 0 \\ \frac{e^{3x} - 1}{x}, & \text{if } x > 0 \end{cases}$

Question 20: Find the intervals in which the function $f(x) = \sin x + \cos x$, where $0 \leq x \leq 2\pi$, is strictly increasing or strictly decreasing.

Question 21: An urn contains 5 red and 5 black balls. A ball is drawn at random, its colour is noted, and it is returned to the urn. Moreover, 2 additional balls of the colour drawn are put in the urn and then a second ball is drawn at random. What is the probability that the second ball is red?

Question 22: Solve for x : $\tan^{-1}\left(\frac{x-1}{x-2}\right) + \tan^{-1}\left(\frac{x+1}{x+2}\right) = \frac{\pi}{4}$.

Question 23: If $y = (\sin^{-1} x)^2$, prove that $(1-x^2)\frac{d^2y}{dx^2} - x\frac{dy}{dx} - 2 = 0$.

Question 24: Using properties of determinants, prove that: $\begin{vmatrix} 1+a & 1 & 1 \\ 1 & 1+b & 1 \\ 1 & 1 & 1+c \end{vmatrix} = abc\left(1 + \frac{1}{a} + \frac{1}{b} + \frac{1}{c}\right)$

SECTION D (15 Marks)

Answer any ONE question from each of the three pairs below.

Question 25: (Option A) Solve the following system of linear equations using the Matrix Method: $2x + 3y + 3z = 5$, $x - 2y + z = -4$, $3x - y - 2z = 3$ **OR (Option B)** Show that the semi-vertical angle of a right circular cone of a given fixed surface area and maximum volume is $\sin^{-1}(1/3)$.

Question 26: (Option A) A diet is to contain at least 80 units of vitamin A and 100 units of minerals. Two foods F_1 and F_2 are available. Food F_1 costs ₹4 per unit and food F_2 costs ₹6 per unit. One unit of food F_1 contains 3 units of vitamin A and 4 units of minerals. One unit of food F_2 contains 6 units of vitamin A and 3 units of minerals. Formulate this as a Linear Programming Problem (LPP) and solve it graphically to find the minimum cost for the diet. (*Note: You must explicitly state all variable definitions and mandatory non-negativity constraints*). **OR (Option B)** Show that the relation R in the set $A = \{x \in \mathbb{Z} : 0 \leq x \leq 12\}$ given by $R = \{(a,b) : |a-b| \text{ is a multiple of } 4\}$ is an equivalence relation. Find the set of all elements related to 1.

Question 27: (Option A) A manufacturer has three machine operators A, B and C. The first operator A produces 1% defective items, whereas the other two operators B and C produce 5% and 7% defective items respectively. A is on the job for 50% of the time, B is on the job for 30% of the time and C is on the job for 20% of the time. A defective item is produced, what is the probability that it was produced by A? **OR (Option B)** Find the area of the greatest rectangle that can be inscribed in an ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.